

Math Lingo: Naming the Parts

Name: _____

Mathematicians have precise names for things, just like grammar does. Today you'll learn to read math the way you read sentences.

Part 1 — Sort It Out

Classify each item. Then explain your reasoning in one short phrase.

Item	Expression, Equation, Formula, or Other?	How do you know?
$3x + 5$		
$7 = 7$		
12		
$4 + 9$		
$y = 3x$		
$A = bh$		
$2(4x^2 - 9) = 8x^2 - 18$		
$x/2 > 6$		

Big question: What is the ONE symbol that separates an expression from an equation?

Part 2 — Expression Anatomy

Use this expression: $8x^2 + 7x - 15$

1. How many terms does it have? List them.

2. What is the coefficient of each term that has a variable?

3. Which term is the constant? How do you know it's still a “term”?

4. Tricky: in the expression $y^2 - y + 3$, what is the coefficient of the middle term?

Now the monster from the slide: $2x^5 - 2x^6y^3z + x^5y^2z^4$

5. How many terms? How many different variables appear in the second term?

6. What is the coefficient of the third term? (Careful — it's invisible!)

Part 3 — Build Your Own

1. Write an expression with exactly 3 terms, where the coefficient of x^2 is 4 and the constant is -7 .

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2. Write an equation that is TRUE but is not a formula.

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3. Write a formula you already know (from area, perimeter, speed, anything!) and say what each letter stands for.

Part 4 — True or False? Defend your answer!

1. Every equation contains two expressions.

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2. 12 is not an expression, because it has no variable.

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3. In $7x$, the 7 is a variable.

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4. In $8x^2 + 7x - 15$, the last term is 15.

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5. Every formula is an equation, but not every equation is a formula.
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Part 5 — Challenge: Read the “Scary” Formula

You will not learn what the quadratic formula DOES for two more years. But you can already READ it:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

1. Which single letter is the variable being solved for?

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2. How many terms are inside the square root, $b^2 - 4ac$?

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3. The whole right-hand side is an expression. Is the entire line an expression or an equation? Why?

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4. Circle every part you can name. Put a ? next to anything you can't name yet — that's your future math!

Part 6 — The Question Compass: Be Your Own Teacher

Strong students don't just answer questions—they invent them. Take ONE idea and ask about it from SIX directions.

Worked example—here is a compass another student filled out for the idea TERM:

Direction	Their questions about TERM (answers in <i>italics</i>)
What	What is a term, in my own words? (<i>A chunk of an expression, separated from other chunks by + or −.</i>)
Example / Non-example	In $8x^2 + 7x - 15$, is $7x$ a term? Is " $8x^2 + 7x$ " a term? (<i>7x yes; the other is TWO terms, so it is not A term—it has a + inside it.</i>)
Reverse	Write an expression with exactly 4 terms where only 2 terms have variables. (<i>Start from the requirements, build backwards: $3x + y + 5 - 2$.</i>)
Break it	Is 12 all by itself a term? Is the minus sign part of the term $-y$ or separate? (<i>Yes — a one-term expression. And the sign travels with the term.</i>)
Connect	How are terms related to coefficients? (<i>Every variable term has a coefficient — even an invisible 1.</i>)
Disguise	How many terms does $2(4x^2 - 9)$ have? (<i>Sneaky: ONE — until you expand it into two.</i>)

Notice: every question is concrete — it points at a real expression, not just the definition. Now you. Today's idea: COEFFICIENT.

Direction	Your question about coefficients
What <i>What is it, in your own words?</i>	
Example / Non-example <i>Give one of each. Why is the non-example not one?</i>	
Reverse <i>Start from an answer, work backwards.</i>	

Break it <i>Find the edge case. When does the idea get weird?</i>	
Connect <i>How does it relate to another idea from today?</i>	
Disguise <i>How would a test HIDE this idea inside a harder problem?</i>	

Now trade compasses. Answer your partner's trickiest question:

Exit Ticket

1. A 5th grader asks you: "What's the difference between an expression and an equation?" Write your answer in 1–2 sentences.

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2. Label ALL the parts you can in: $5y^2 - y + 3 = 0$ (terms, coefficients, variable, constant, expression, equation)
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